

Problem statement

The rapid adoption of IoT devices has increased connectivity but also brought about several cybersecurity challenges. Most IoT devices are not well protected which makes them a soft target for any malicious or abnormal network activity. Traditional intrusion detection systems normally rely on labelled attack data. Therefore, it is hard for them to recognize new or unknown threats. This project investigates whether a deep learning model, more specifically an autoencoder, can learn normal IoT network behaviour in a way that automatically detects unusual activity or anomalies without relying labelled datasets.

Aims and objectives

Aims

To analyze the challenges regarding security within IoT networks and understand how unusual or suspicious activities affect IOT devices. Design and develop a data driven detection system capable of detecting unusual activities in IoT traffic by recognizing deviations from normal network behaviour. The performance and practicality of the proposed system would be evaluated for the detection of unusual activity in various IOT network scenarios.

Objectives

Do background research into the IOT security risks, network vulnerabilities and existing methods for detecting abnormal activity. Compare different traditional intrusion detection systems and review the limitations of each with respect to new unseen attacks. Collect and prepare the dataset of IOT network traffic such as BoT-IoT and TON_IoT. Perform data cleaning, scaling and feature selection. Develop a model or algorithm for detection that can learn what normal IoT network activity looks like and identify deviations or anomalies from it. The system should be tested and evaluated based on a variety of performance metrics including accuracy, recall and false positive rate. Compare the outcome against simpler or existing detection methods to measure improvement in detecting unusual network behavior.

Project title and author

Using autoencoders to detect unusual activity in IOT networks
By Mohamed Ibrahim



Legal, ethical and professional

Legal responsiblities refer to the legal and regulatory requirements about your project. It could be about data protection laws, such as GDPR, when personal or sensitive is handled. Such situations would include rightds over intellectual properly correctly using licensed datasets or open sourced software. Cybersecurity regulations will make sure that your system does not untentionally enable harm or a breach. Ethics are about doing the right thing, beyond what is legally necessary, based on ethical values. Key ethical issues are privacy: handling sensitive data securely and refraining from misuse of user information. Trasparency: honesty about the capabilities and limitations of the sysrem should be implemented. Bias and Fairness: models or systems should not discriminate or treat users or data types inequitably. Informed consent: participants shoud understand and agree to the handling of their data if it involves human data.

Professionalism means acting responsibly and demonstratring integrity in your work. This includes the following: Following industry norms and standards in terms of best practices, such as secure coding and testing protocols. Ensuring clear documentation and communication so others can understand, use, and extend your work. Accountability, which means taking ownership of contributions and issues arising from this. Respecting confidentiality when dealing with sensitive or proprietary information.

Key literature/ references

[1] M. A. Al-Garadi, A. Mohamed, A. K. Al-Ali, X. Du, I. Ali, and M. Guizani, “A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security,” *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1646–1685, Apr. 2020, doi: <https://doi.org/10.1109/comst.2020.2988293>.

[2] Y. Meidan *et al.*, “N-BaloT—Network-Based Detection of IoT Botnet Attacks Using Deep Autoencoders,” *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 12–22, Jul. 2018, doi: <https://doi.org/10.1109/mprv.2018.03367731>.

[3] K. Dubey, R. Dubey, Sudhakar Panedy, and S. Kumar, “A Review of IoT Security: Machine Learning and Deep Learning Perspective,” *Procedia Computer Science*, vol. 235, pp. 335–346, Jan. 2024, doi: <https://doi.org/10.1016/j.procs.2024.04.034>.

Use of AI

The project will use artificial intelligence through a machine learning model called an autoencoder, which learns the patterns of normal IoT network activity and detects unusual behaviour. I will process and validate the model's output by comparing the detected anomalies with the original dataset and the known patterns of the network traffic to ensure the realism of the results. Any risks of bias or inaccuracy will be minimized by using diverse, publicly available IoT datasets, careful data preprocessing, and the use of multiple performance metircs in evaluation to confirm the reliability and accuracy of the system under different scenarios.